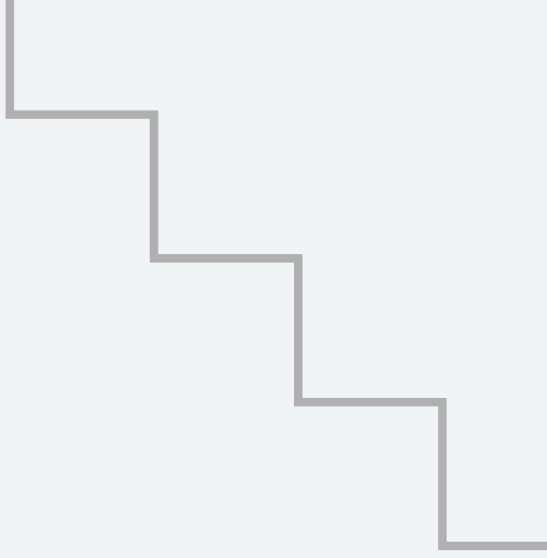
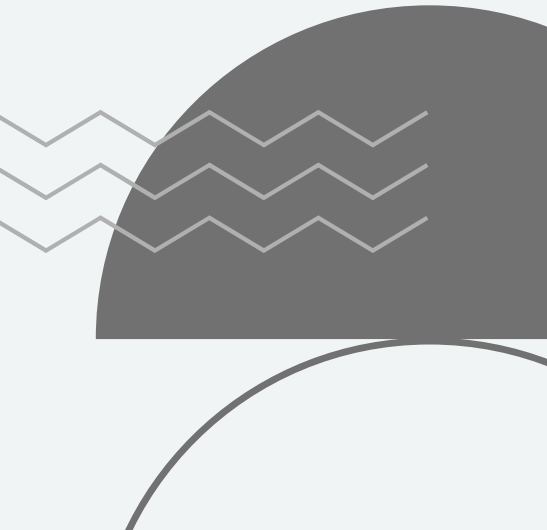
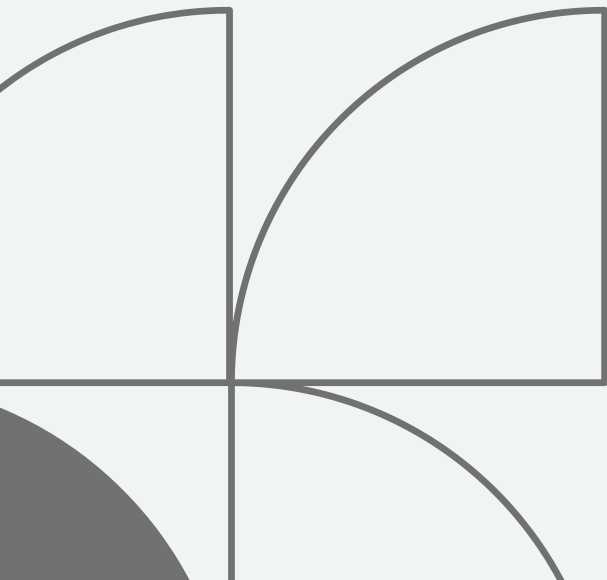
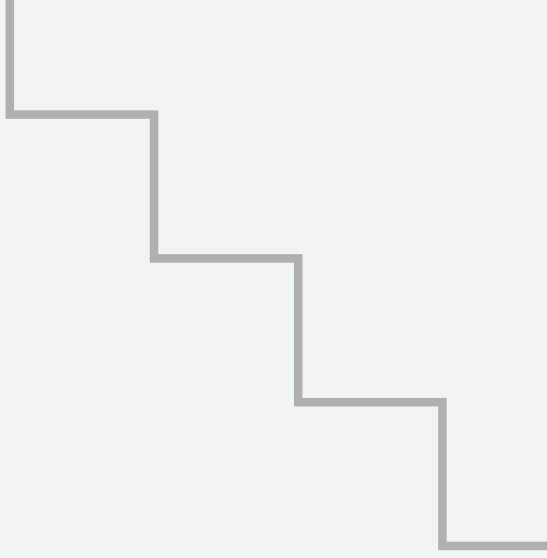




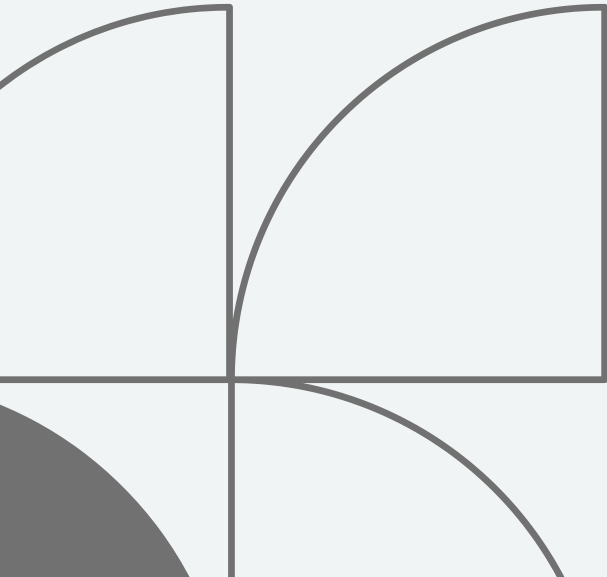
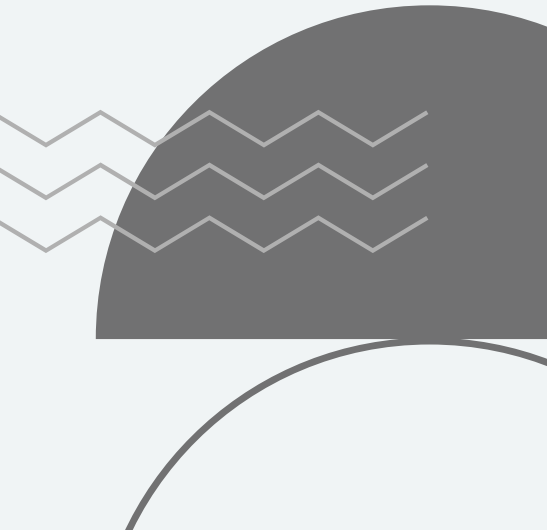
DIGITAL APPROVAL SYSTEM

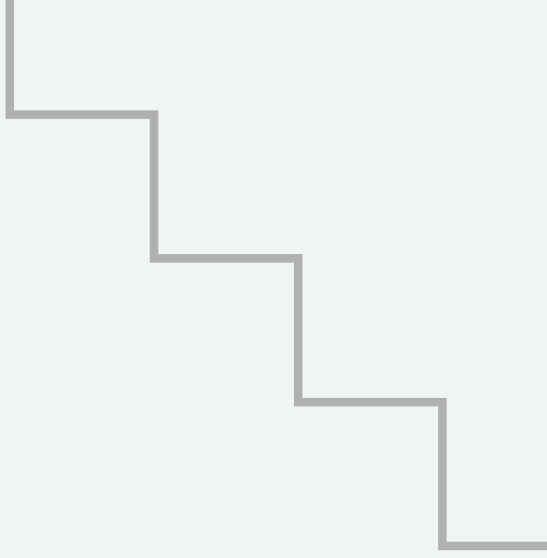
By Group 5



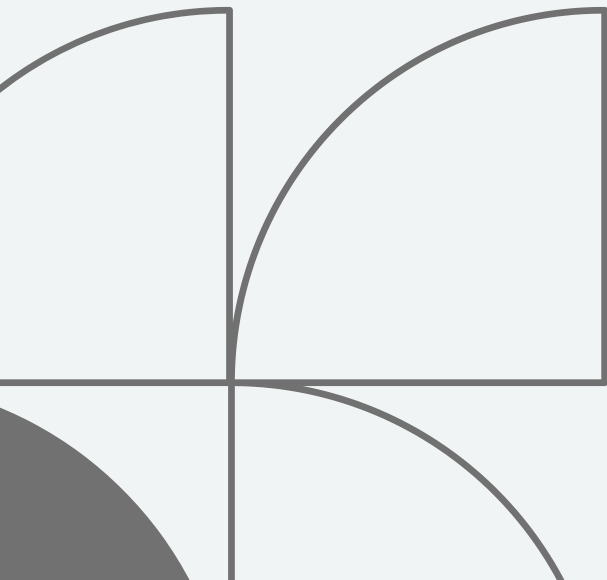
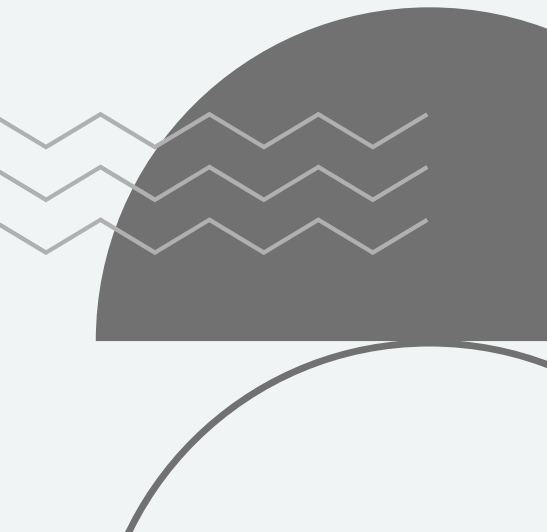


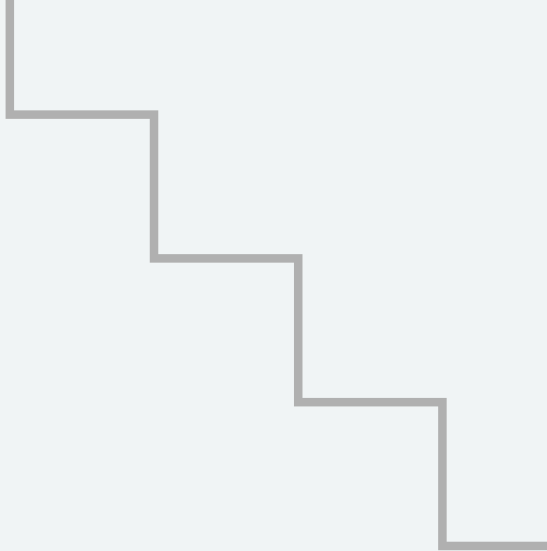
INTRODUCTION

- Manual paper approval involves students moving from tutor to HOD to principal for signatures.
 - This causes delay and inconvenience.
 - Our system digitizes this process to make approval fast, trackable, and paperless.
 - Developed entirely using Java Swing GUI and file handling .
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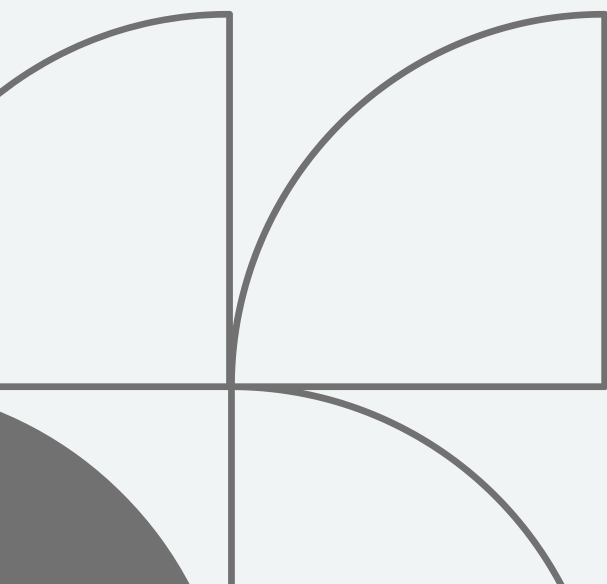
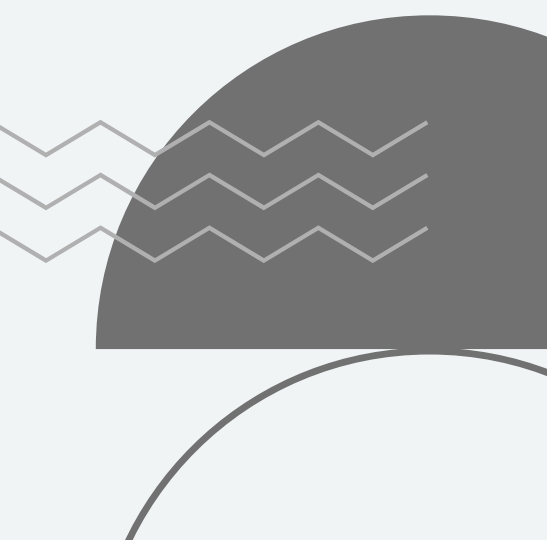


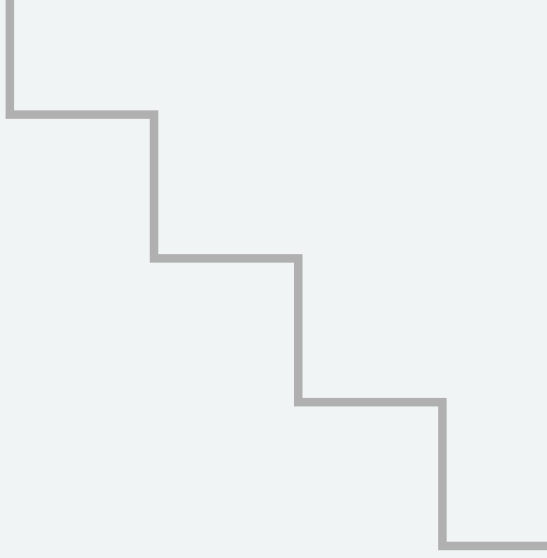
OBJECTIVE OF THE PROJECT

- To reduce time and effort in paper approvals.
 - To automate the submission and review workflow.
 - To ensure transparency and easy tracking of approval status.
 - To provide separate logins for students, tutors, HOD, deans, and principal.
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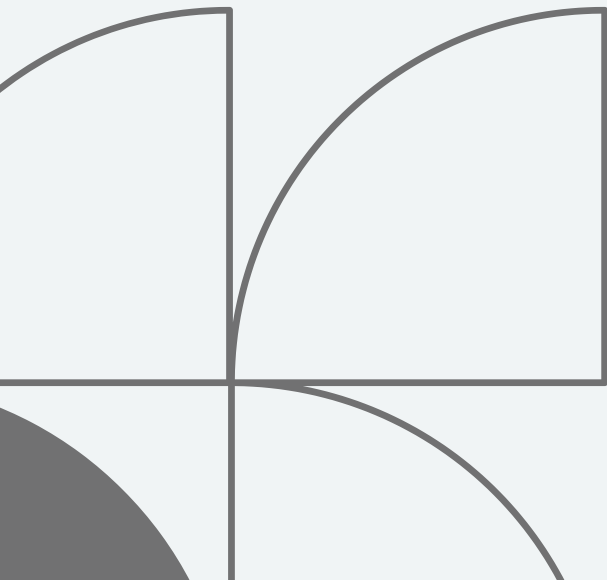
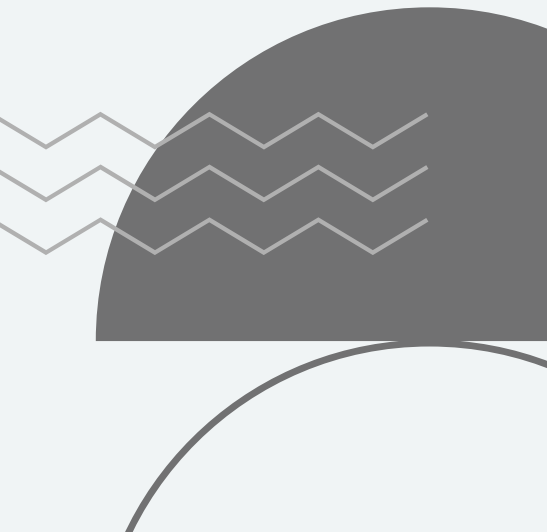


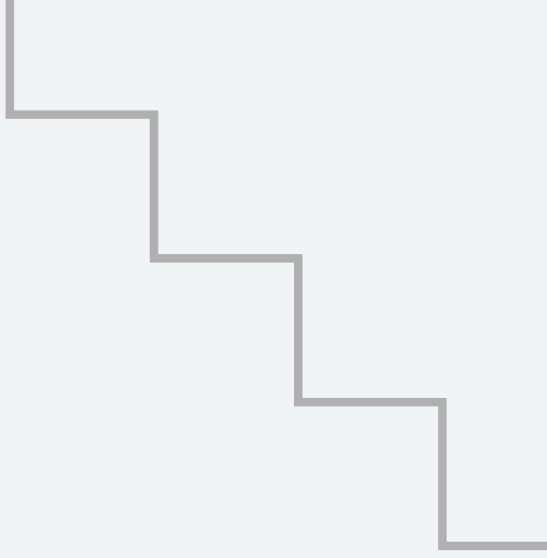
PROBLEM WITH EXISTING MANUAL SYSTEM

- Students must find tutors and HODs in person.
 - Delays due to teacher unavailability.
 - Misplacement or damage of papers.
 - No central record of approval status.
- 
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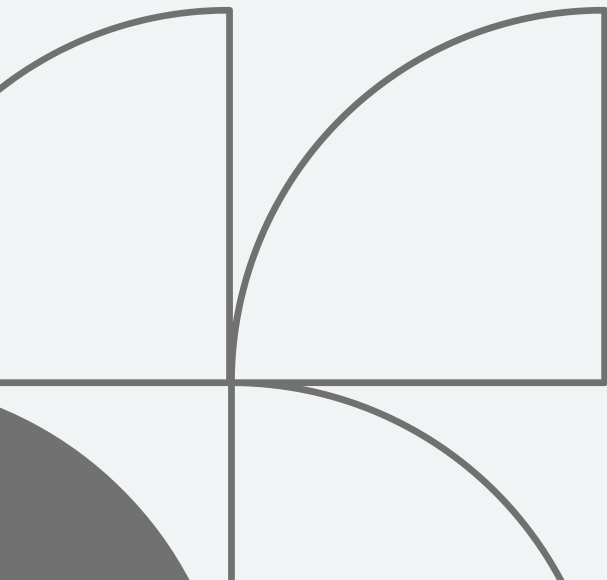
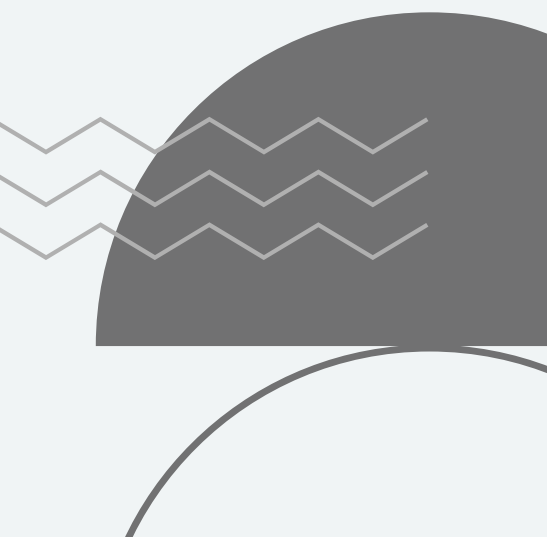


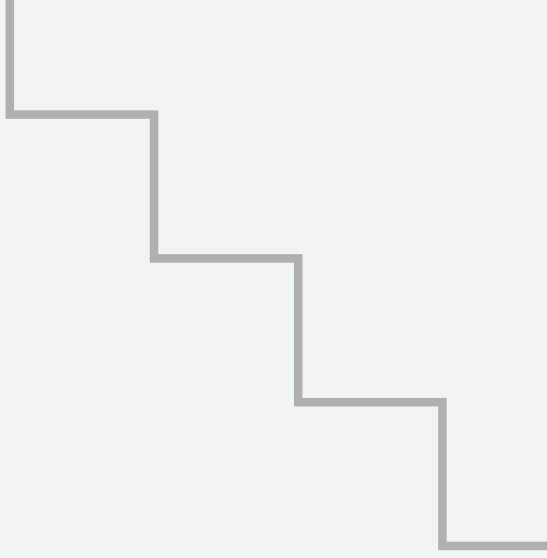
PROPOSED APPLICATION

- Students submit papers online.
 - Automatically routed to assigned tutor, then HOD, dean, and principal.
 - Faculty can approve/reject digitally.
 - All approvals are stored and can be tracked easily.
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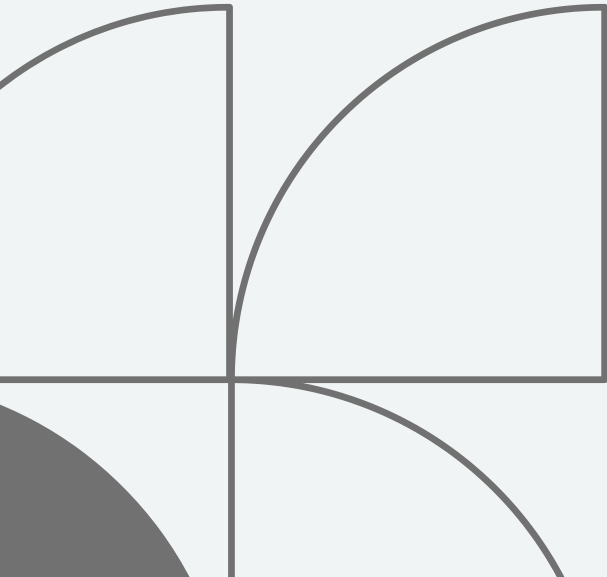
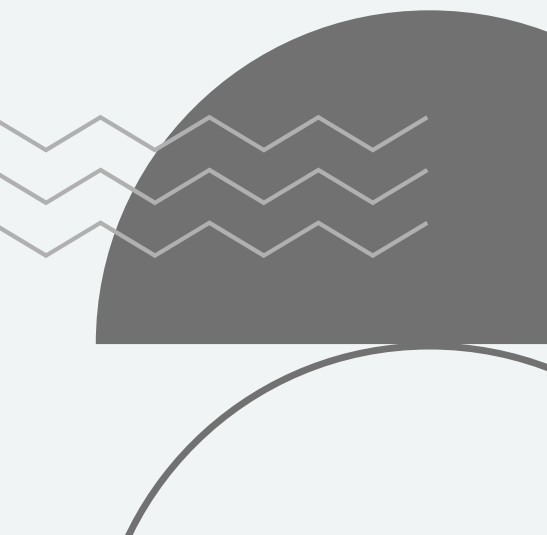


SYSTEM ARCHITECTURE

- Frontend: Java Swing GUI for all user interfaces.
 - Backend: Java File Handling for data storage.
 - Users: Students, Tutors, HODs, Deans, Principal.
 - Flow:
Student → Tutor → HOD → Dean (if needed) → Principal
- 
- 

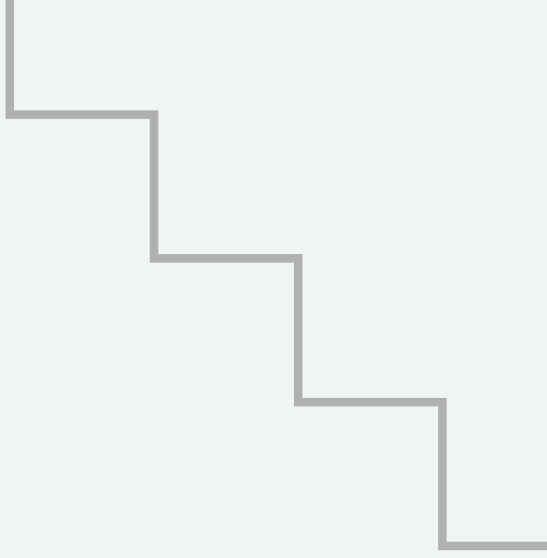


TECHNOLOGIES USED:

- Programming Language: Java
 - IDE: Visual Studio Code
 - Frontend: Java Swing (JFrame, JPanel, JButton, JLabel, JTextField, JTable, JOptionPane)
 - Backend: File Handling using FileReader, FileWriter, BufferedReader, BufferedWriter
 - Storage: .txt files for user data, submissions, and approvals
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MODULES IMPLEMENTED:



1.Login & Registration Module

- Separate registration for students and faculty.
- Predefined users (HODs, tutors, principal, dean).

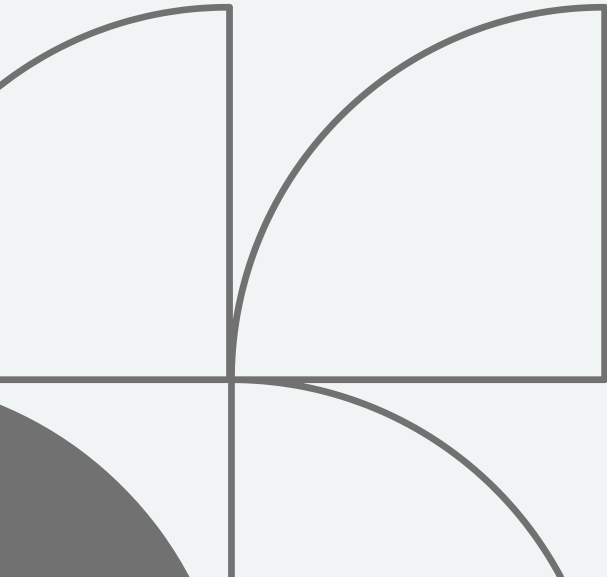
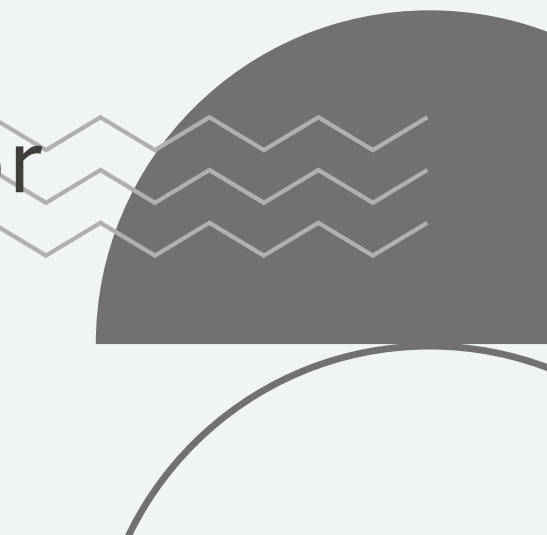
2.Paper Submission Module

- Students upload and submit papers digitally.

3.Approval Module

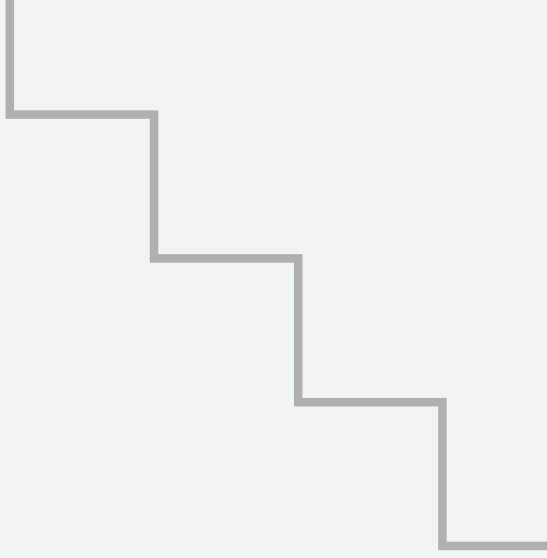
- Tutor, HOD, Dean, and Principal can approve/reject.

4.Status Tracking Module

- Shows real-time status like “Pending at Tutor” or “Approved by HOD.”
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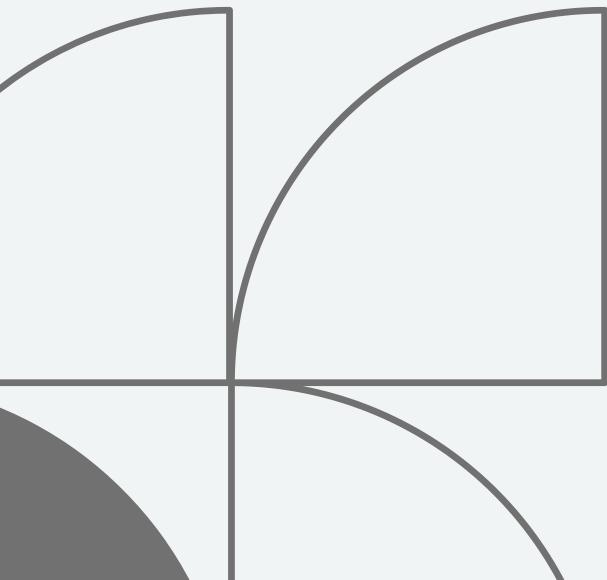
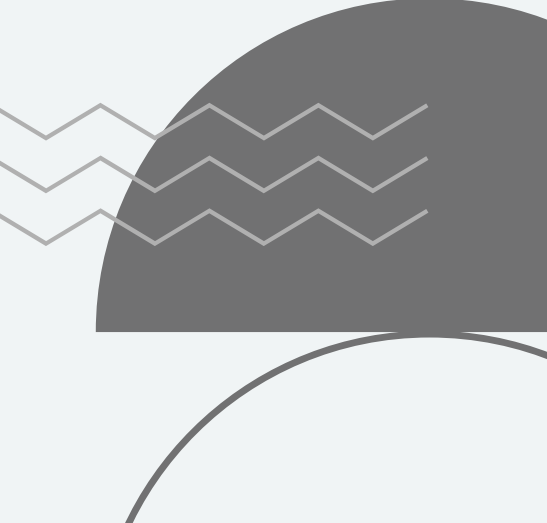
SWING COMPONENTS USED:

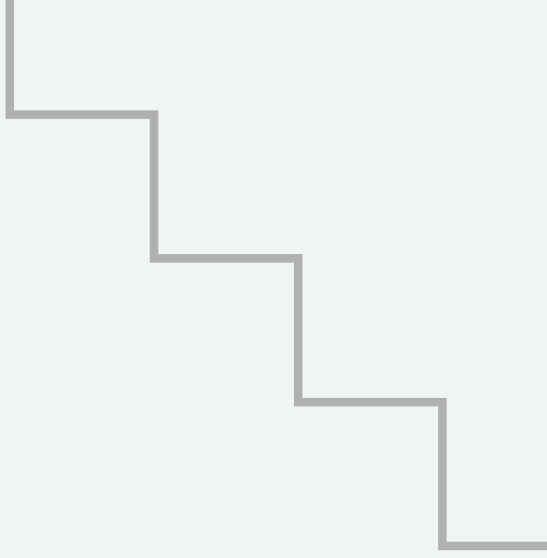
Component	Description	Example in Your Code
JFrame	The main window of the app.	JFrame frame = new JFrame("File Editor");
JPanel	A container to hold components.	Used for layout organization.
JButton	Clickable buttons.	“Open”, “Save”, “New File”, “Exit”.
JTextArea	Area to type and display text.	Used to show and edit file content.
JScrollPane	Adds scrolling to the text area.	Wrapped around JTextArea.
JLabel	Displays small text labels.	(If used) to show messages.
JFileChooser	Opens a dialog box to select files.	Used for “Open” and “Save As” functions.
ActionListener	Interface to handle button clicks.	Used to perform tasks when a button is clicked.
BorderLayout, FlowLayout	Layout managers that decide how components appear.	Used implicitly in your code.



CLASS DIAGRAM

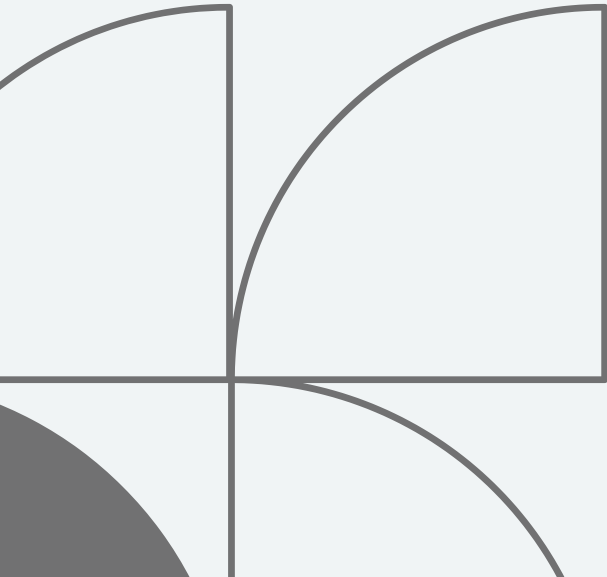
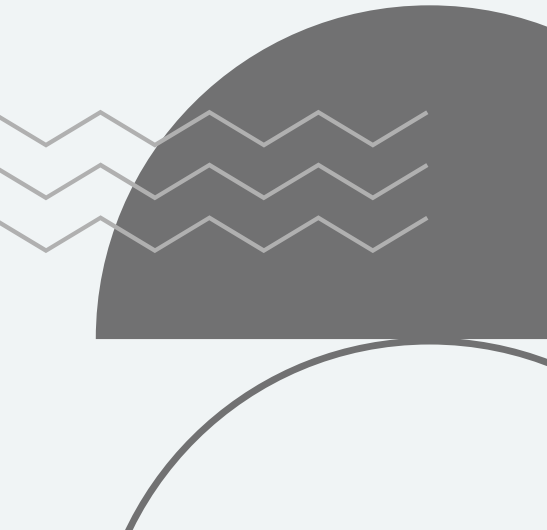
Main Classes:

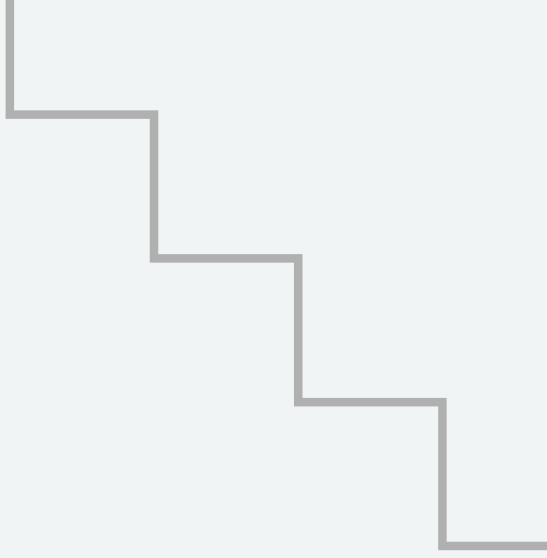
- User – Common superclass for all users.
 - Student – Handles paper submission and status.
 - Faculty – For tutors, HODs, deans, and principal.
 - Document – Stores paper details and approval status.
 - DigitalApprovalSystem – Main control class with GUI and logic.
- 
- 



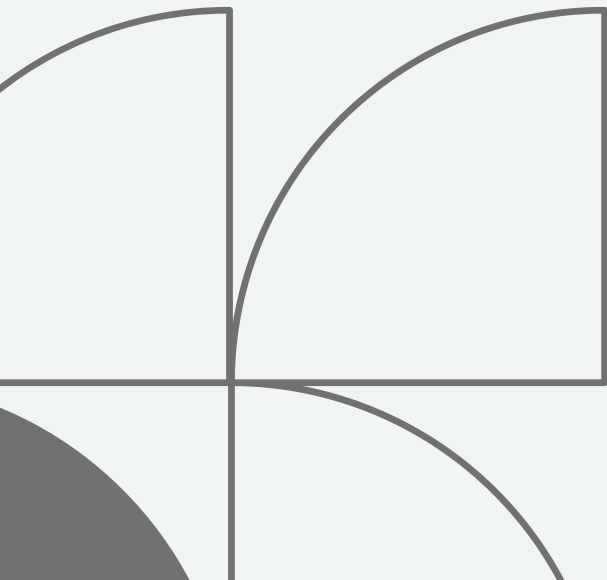
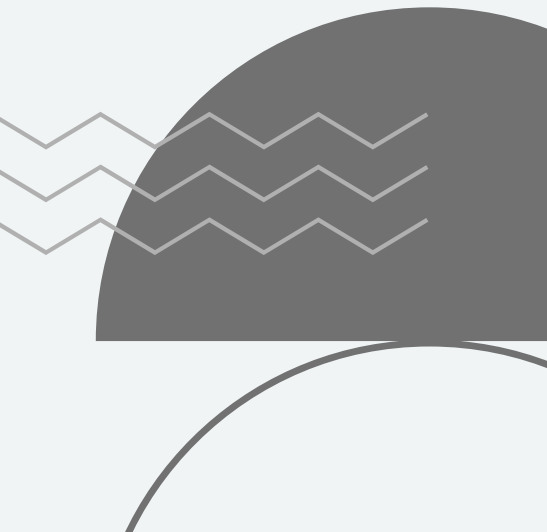
CLASS DIAGRAM

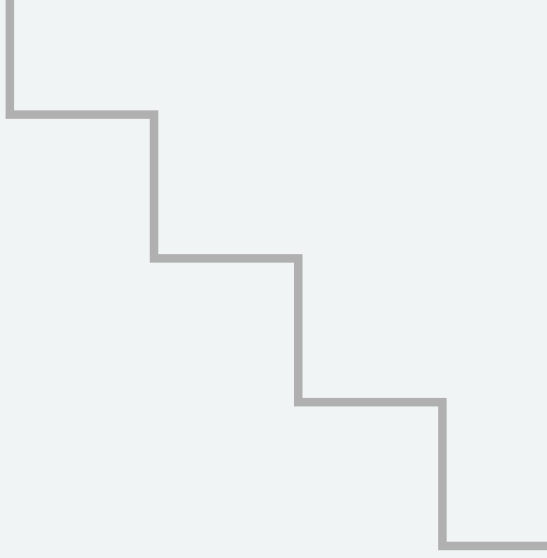
Relationships:

- Student → submits Document.
 - Faculty → reviews Document.
 - DigitalApprovalSystem → coordinates all interactions.
- 
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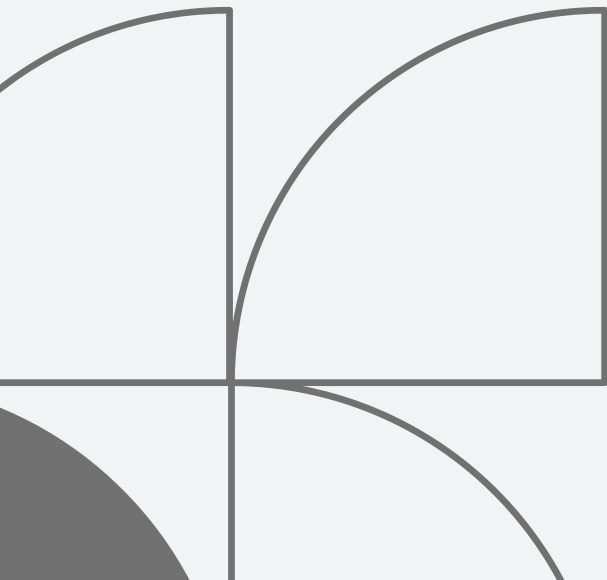
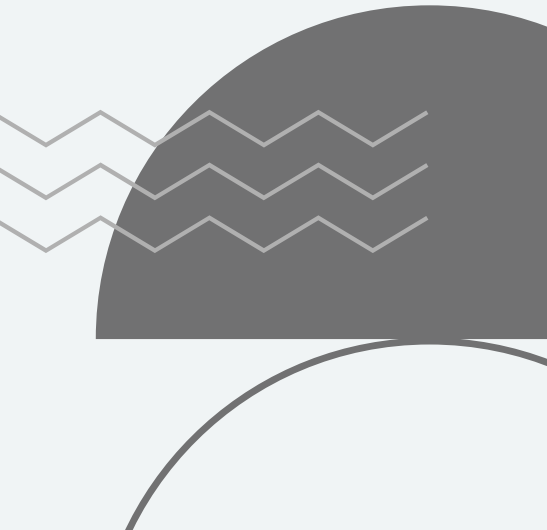
FUNCTIONAL FLOW

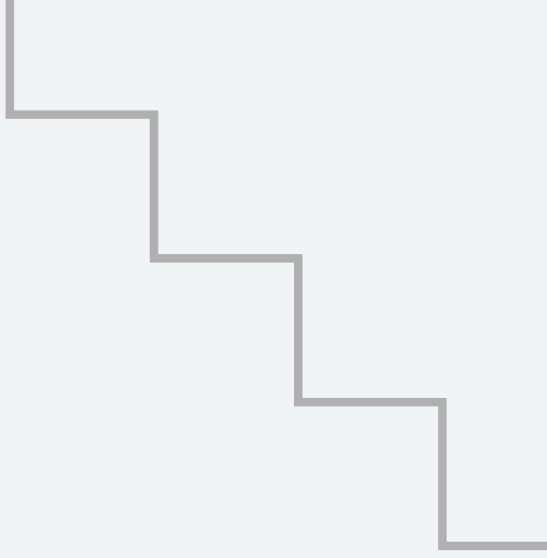
1. User logs in or registers.
 2. Student selects department, year, and class.
 3. System assigns correct tutor and HOD automatically.
 4. Student submits document.
 5. Document moves to tutor → HOD → Dean → Principal sequentially.
 6. Status updates at every level.
- 
- 



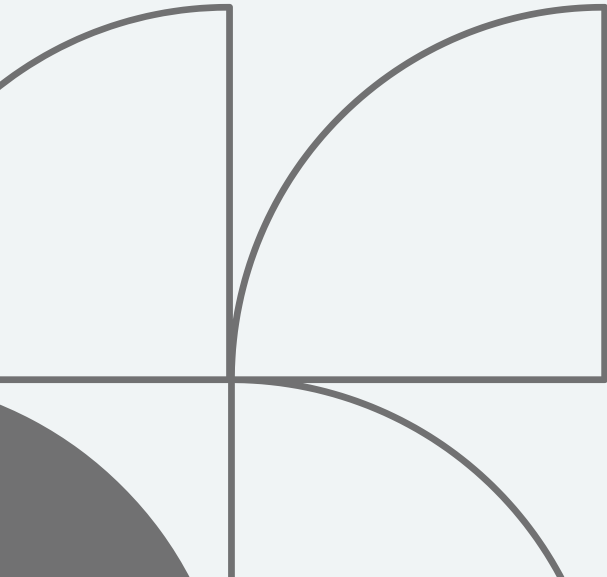
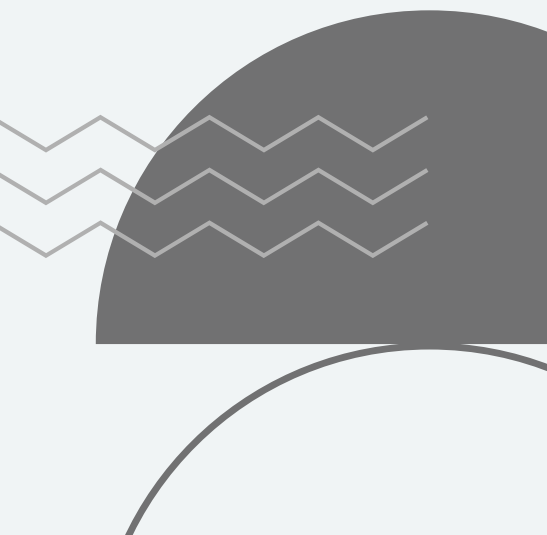
USER INTERFACE OVERVIEW

Swing Components Used:

- JFrame, JPanel – Window & layout.
 - JButton, JLabel, JTextField – Inputs and actions.
 - JComboBox – For class/year/department selection.
 - JOptionPane – Display pop-up messages.
 - JTable – Display list of pending papers.
- 
- 



FILE HANDLING IN THE SYSTEM

- All data (login info, documents, approvals) stored in .txt files.
 - Files:
 - users.txt – user credentials
 - submissions.txt – student submissions
 - status.txt – current approval statuses
 - File Handling Classes Used:
 - FileWriter, FileReader
 - BufferedWriter, BufferedReader
- 
- 

TEST CASES

Test Case ID	Scenario	Input	Expected Output
TC01	Student login	Valid username/password	Dashboard opens
TC02	Student registration	New user data	Account created
TC03	Submit paper	File upload	“Submitted to Tutor”
TC04	Tutor approval	Click Approve	Status → Approved
TC05	Invalid login	Wrong password	Error message

STUDENT LOGIN

Digital Approval System - Login

Digital Approval System

Username:

Password:

Digital Approval System - Login

Digital Approval System

Username:

Password:

STUDENT REGISTRATION

Student Registration

Full Name:

Aysha Arshad Hussain

Password:

.....

Department:

CSE

Year:

2

Class:

CSE B (2)

Create Student

Back

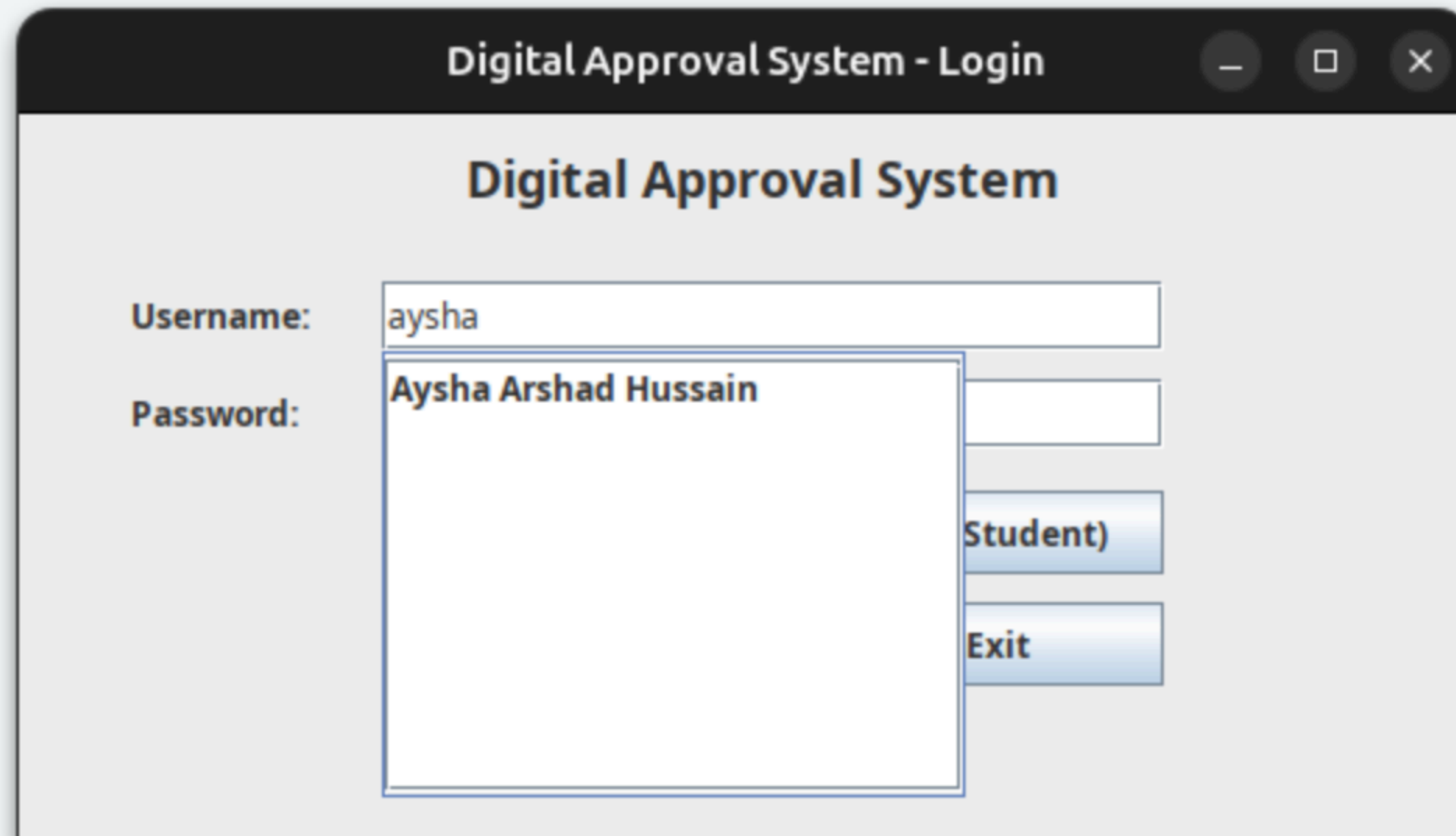
Message

i

Student account created. Your username is: ayshaarshadhussain

OK

INVALID LOGIN



The screenshot shows a login window titled "Digital Approval System - Login". The window has a title bar with standard minimize, maximize, and close buttons. The main content area has a title "Digital Approval System" and two input fields: "Username:" and "Password:". The "Username:" field contains the text "aysha". The "Password:" field contains the text "Aysha Arshad Hussain". Below the password field, there is a dropdown menu that is open, showing a list of names. The first name in the list is "Aysha Arshad Hussain". To the right of the password field, there are two buttons: "Student)" and "Exit".

Digital Approval System - Login

Digital Approval System

Username:

Password:

Aysha Arshad Hussain

Student)

Exit

PAPER SUBMISSION

Dashboard - Digital Approval System

Logged in: Aysha Arshad Hussain [ayshaarshadhussain] - Student

ProfileLogout

Submit PaperRefresh

ID	Title	Dept	Class	Status	History
P1761117078671	Retest Request	CSE	CSE B (2)	Pending (Tutor)	Submitted by Aysha Arshad H..

Input

?

Enter paper title:

Retest Request

OKCancel

PAPER SUBMISSION

Dashboard - Digital Approval System

Logged in: Aysha Arshad Hussain [ayshaarshadhussain] - Student

ProfileLogout

Submit PaperRefresh

ID	Title	Dept	Class	Status	History
P1761117078671	Retest Request	CSE	CSE B (2)	Pending (Tutor)	Submitted by Aysha Arshad H.

Enter paper content:

?

Subject: Request for Retest Permission

Respected Ma'am,

I am Aysha Arshad Hussain, studying in S3 CSE B, Roll No. 17.

I was unable to attend the series test for THEORY OF COMPUTATION(PCCST).

I sincerely apologize for my absence and kindly request your consideration in

OKCancel

PAPER SUBMISSION

Dashboard - Digital Approval System

Logged in: Aysha Arshad Hussain [ayshaarshadhussain] - Student

Profile

Logout

Submit Paper

Refresh

ID	Title	Dept	Class	Status	History
----	-------	------	-------	--------	---------

Next Approver

?

Choose who should review next (default Tutor):

Tutor

OK

Cancel

PAPER SUBMISSION

Dashboard - Digital Approval System

Logged in: Aysha Arshad Hussain [ayshaarshadhussain] - Student

Profile

Logout

Submit Paper

Refresh

ID	Title	Dept	Class	Status	History
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Message

i

Sent to Prof. Vengayil Nayana Murali for approval.

OK

PAPER APPROVAL

Dashboard - Digital Approval System					
Logged in: Prof. Vengayil Nayana Murali [vengayilnayanamurali] - Tutor					Profile Logout
Refresh View Relevant					
ID	Title	SubmittedBy	Dept	Class	Status
P1759817248460	sample	Devangana S S	CSE	CSE B (2)	Rejected by Tutor
P1761117078671	Retest Request	Aysha Arshad Hussain	CSE	CSE B (2)	Pending (Tutor)

PAPER APPROVAL

Dashboard - Digital Approval System

Logged in: Prof. Vengayil Nayana Murali [vengayilnayanamurali] - Tutor

Profile

Logout

Refresh

View Relevant

ID	Title	SubmittedBy	Dept	Class	Status
P1759817248460	sample	Devangana S S	CSE	CSE B (2)	Rejected by Tutor
P1761117078671	Retest Request	Aysha Arshad Hussain	CSE	CSE B (2)	Pending (Tutor)

Approve

Reject

Details

PAPER APPROVAL

Dashboard - Digital Approval System

Logged in: Prof. Vengayil Nayana Murali [vengayilnayanamurali] - Tutor

Profile

Logout

Refresh

View Relevant

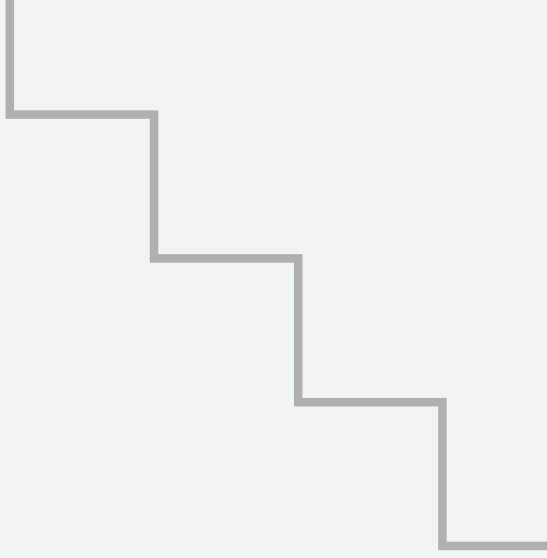
ID	Title	SubmittedBy	Dept	Class	Status
P1759817248460	sample	Devangana S S	CSE	CSE B (2)	Rejected by Tutor
P1761117078671	Retest Request	Aysha Arshad Hussain	CSE	CSE B (2)	Pending (Tutor)

Message

i

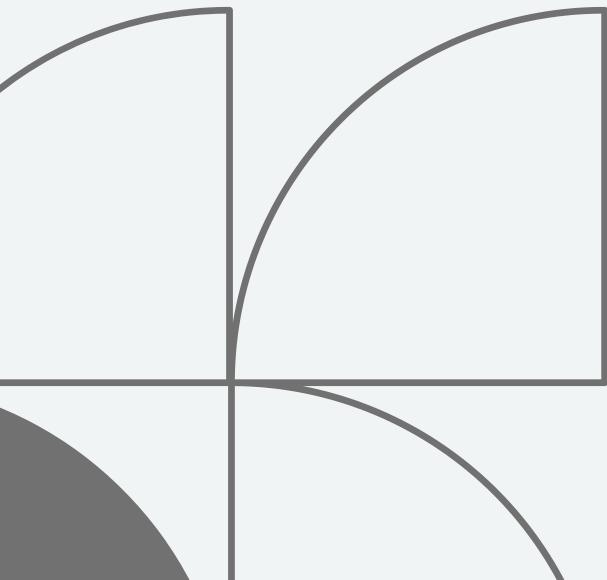
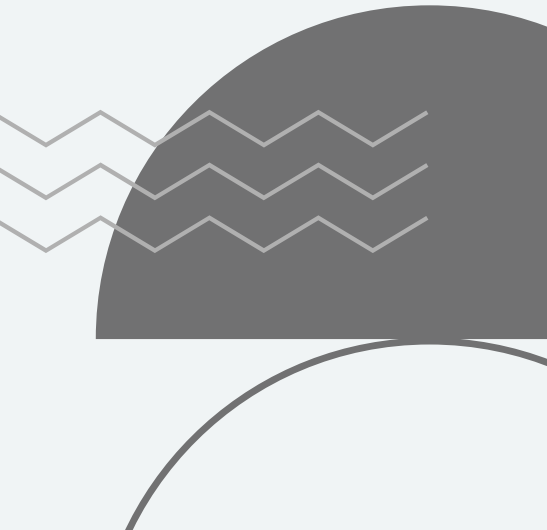
Approved. Next: HOD

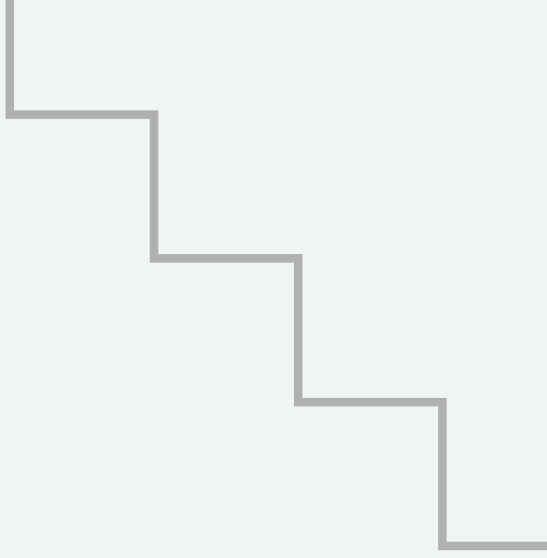
OK



ADVANTAGES & LIMITATIONS

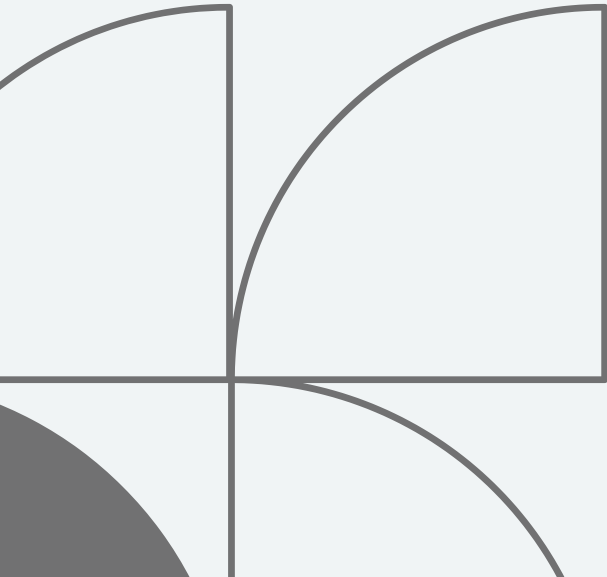
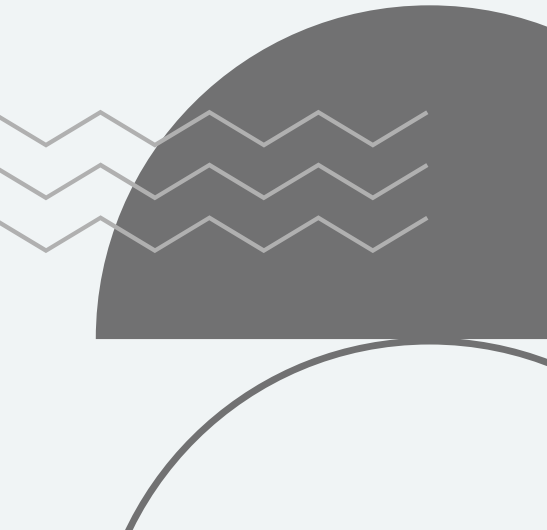
Advantages:

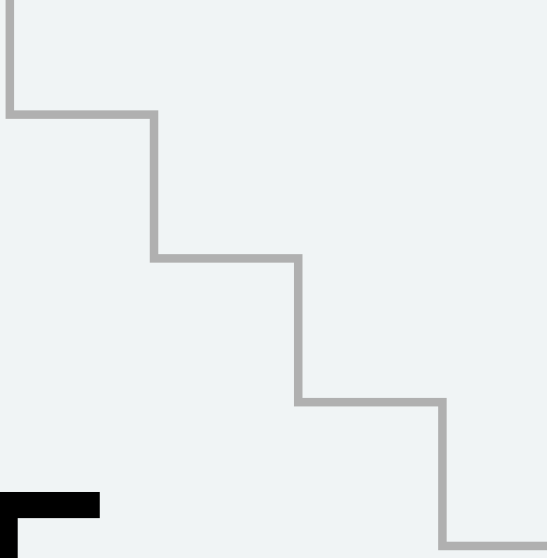
- Eliminates paper movement.
 - Saves time and effort.
 - Real-time status tracking.
 - Accessible from anywhere.
- 
- 



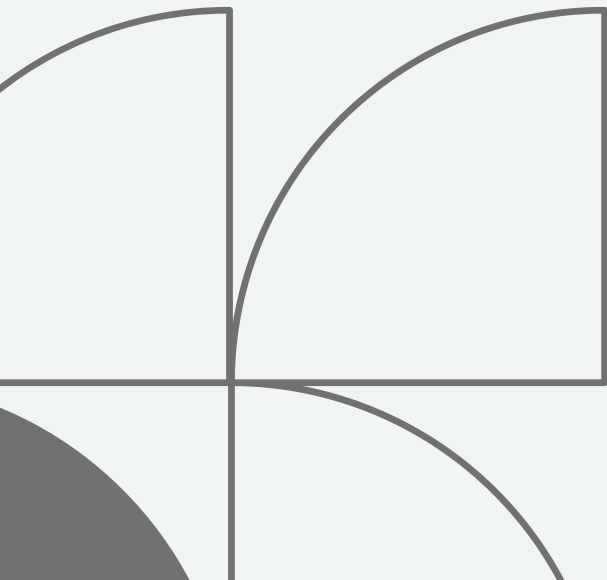
ADVANTAGES & LIMITATIONS

Limitations:

- Works on a single system (local file-based).
 - No cloud or database support yet.
 - Limited to GUI-based environment.
- 
- 



CONCLUSION & FUTURE SCOPE

- Digitizes the entire paper approval process.
 - Reduces manual effort and human error.
 - Future Enhancements:
 - Database integration (MySQL).
 - Role-based authentication.
 - Email notifications.
 - Web-based deployment.
- 
- 